

Tutorial: Custom YOLO Object Detection with RoboFlow

In this tutorial, you'll learn how to create a custom YOLO object detection model using your own classes and annotations with RoboFlow.

Step 1: Install Required Dependencies

```
%pip install --upgrade ultralytics
%pip install opencv-python
%pip install roboflow
%pip install matplotlib
%pip install numpy==1.26.4 --force-reinstall
```

Collecting ultralytics

Downloading ultralytics-8.4.26-py3-none-any.whl.metadata (39 kB)

Requirement already satisfied: numpy>=1.23.0 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ultralytics) (1.26.4)

Collecting matplotlib>=3.3.0 (from ultralytics)

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Requirement already satisfied: opencv-python>=4.6.0 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ultralytics) (4.13.0.92)

Requirement already satisfied: pillow>=7.1.2 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ultralytics) (12.1.1)

Requirement already satisfied: pyyaml>=5.3.1 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ultralytics) (6.0.3)

Requirement already satisfied: requests>=2.23.0 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ultralytics) (2.32.5)

Requirement already satisfied: scipy>=1.4.1 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ultralytics) (1.17.1)

Collecting torch>=1.8.0 (from ultralytics)

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Collecting torchvision>=0.9.0 (from ultralytics)

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Requirement already satisfied: psutil>=5.8.0 in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from ultralytics) (7.2.2)

Collecting polars>=0.20.0 (from ultralytics)

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Requirement already satisfied: kiwisolver>=1.3.1 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib>=3.3.0->ultralytics) (1.5.0)
Requirement already satisfied: packaging>=20.0 in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from matplotlib>=3.3.0->ultralytics) (26.0)
Requirement already satisfied: pyparsing>=3 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib>=3.3.0->ultralytics) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from matplotlib>=3.3.0->ultralytics) (2.9.0.post0)
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Requirement already satisfied: six>=1.5 in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from python-dateutil>=2.7->matplotlib>=3.3.0->ultralytics) (1.17.0)
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Requirement already satisfied: certifi>=2017.4.17 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from requests>=2.23.0->ultralytics) (2026.2.25)
Collecting filelock (from torch>=1.8.0->ultralytics)
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Requirement already satisfied: typing-extensions>=4.10.0 in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from

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torch>=1.8.0->ultralitics) (65.5.0)
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Collecting mpmath<1.4,>=1.1.0 (from sympy>=1.13.3->torch>=1.8.0-
>ultralitics)
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Downloading filelock-3.25.2-py3-none-any.whl (26 kB)
Using cached jinja2-3.1.6-py3-none-any.whl (134 kB)
Installing collected packages: mpmath, sympy, polars-runtime-32,
numpy, networkx, jinja2, fsspec, filelock, torch, polars, ultralytics-
thop, torchvision, matplotlib, ultralytics
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Successfully installed filelock-3.25.2 fsspec-2026.2.0 jinja2-3.1.6
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1.39.3 polars-runtime-32-1.39.3 sympy-1.14.0 torch-2.11.0 torchvision-

0.26.0 ultralytics-8.4.26 ultralytics-thop-2.0.18

Note: you may need to restart the kernel to use updated packages.

WARNING: Ignoring invalid distribution ~atplotlib (c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages)

WARNING: Ignoring invalid distribution ~atplotlib (c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages)

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

scikit-learn 1.3.2 requires numpy<2.0,>=1.17.3, but you have numpy 2.4.3 which is incompatible.

tensorflow 2.21.0 requires keras>=3.12.0, but you have keras 2.15.0 which is incompatible.

tensorflow 2.21.0 requires ml_dtypes<1.0.0,>=0.5.1, but you have ml_dtypes 0.2.0 which is incompatible.

tensorflow 2.21.0 requires protobuf<8.0.0,>=6.31.1, but you have protobuf 4.25.8 which is incompatible.

WARNING: Ignoring invalid distribution ~atplotlib (c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages)

Requirement already satisfied: opencv-python in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (4.13.0.92)

Requirement already satisfied: numpy>=2 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from opencv-python) (2.4.3)

Note: you may need to restart the kernel to use updated packages.

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WARNING: Ignoring invalid distribution ~atplotlib (c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages)

Collecting roboflow

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Requirement already satisfied: certifi in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (2026.2.25)

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Requirement already satisfied: cyclical in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (0.12.1)

Requirement already satisfied: kiwisolver>=1.3.1 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (1.5.0)

Requirement already satisfied: matplotlib in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from

roboflow) (3.10.8)
Requirement already satisfied: numpy>=1.18.5 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (2.4.3)
Collecting opencv-python-headless==4.10.0.84 (from roboflow)
 Downloading opencv_python_headless-4.10.0.84-cp37-abi3-win_amd64.whl.metadata (20 kB)
Requirement already satisfied: Pillow>=7.1.2 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (12.1.1)
Collecting pillow-avif-plugin<2 (from roboflow)
 Downloading pillow_avif_plugin-1.5.5-cp311-cp311-win_amd64.whl.metadata (2.3 kB)
Requirement already satisfied: python-dateutil in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from roboflow) (2.9.0.post0)
Collecting python-dotenv (from roboflow)
 Downloading python_dotenv-1.2.2-py3-none-any.whl.metadata (27 kB)
Requirement already satisfied: requests in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (2.32.5)
Requirement already satisfied: six in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from roboflow) (1.17.0)
Requirement already satisfied: urllib3>=1.26.6 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (2.6.3)
Collecting tqdm>=4.41.0 (from roboflow)
 Downloading tqdm-4.67.3-py3-none-any.whl.metadata (57 kB)
Requirement already satisfied: PyYAML>=5.3.1 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (6.0.3)
Collecting requests-toolbelt (from roboflow)
 Downloading requests_toolbelt-1.0.0-py2.py3-none-any.whl.metadata (14 kB)
Collecting filetype (from roboflow)
 Downloading filetype-1.2.0-py2.py3-none-any.whl.metadata (6.5 kB)
Collecting pi-heif<2 (from roboflow)
 Downloading pi_heif-1.3.0-cp311-cp311-win_amd64.whl.metadata (6.5 kB)
Requirement already satisfied: colorama in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from tqdm>=4.41.0->roboflow) (0.4.6)
Requirement already satisfied: contourpy>=1.0.1 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib->roboflow) (1.3.3)
Requirement already satisfied: fonttools>=4.22.0 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib->roboflow) (4.62.1)
Requirement already satisfied: packaging>=20.0 in C:\Users\cyanezru\

```

AppData\Roaming\Python\Python311\site-packages (from matplotlib-
>roboflow) (26.0)
Requirement already satisfied: pyparsing>=3 in c:\Users\cyanezru\
AppData\Local\Programs\Python\Python311\Lib\site-packages (from
matplotlib->roboflow) (3.3.2)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\Users\
cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages
(from requests->roboflow) (3.4.6)
Downloading roboflow-1.2.16-py3-none-any.whl (95 kB)
Downloading idna-3.7-py3-none-any.whl (66 kB)
Downloading opencv_python_headless-4.10.0.84-cp37-abi3-win_amd64.whl
(38.8 MB)
----- 0.0/38.8 MB ? eta -:--:--
----- 1.3/38.8 MB 6.1 MB/s eta
0:00:07
----- 3.4/38.8 MB 8.4 MB/s eta
0:00:05
----- 6.3/38.8 MB 10.2 MB/s eta
0:00:04
----- 7.9/38.8 MB 10.6 MB/s eta
0:00:03
----- 8.7/38.8 MB 8.4 MB/s eta
0:00:04
----- 9.4/38.8 MB 7.5 MB/s eta
0:00:04
----- 12.1/38.8 MB 8.2 MB/s eta
0:00:04
----- 15.7/38.8 MB 9.3 MB/s eta
0:00:03
----- 18.4/38.8 MB 9.6 MB/s eta
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----- 20.2/38.8 MB 9.4 MB/s eta
0:00:02
----- 22.0/38.8 MB 9.3 MB/s eta
0:00:02
----- 24.6/38.8 MB 9.6 MB/s eta
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----- 27.8/38.8 MB 10.0 MB/s eta
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----- 29.4/38.8 MB 9.8 MB/s eta
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----- 30.4/38.8 MB 9.6 MB/s eta
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----- 32.0/38.8 MB 9.4 MB/s eta
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----- 32.5/38.8 MB 8.9 MB/s eta
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----- 33.6/38.8 MB 8.9 MB/s eta
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```

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----- 34.9/38.8 MB 8.6 MB/s eta
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----- 36.2/38.8 MB 8.2 MB/s eta
0:00:01
----- 37.0/38.8 MB 7.9 MB/s eta
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----- 38.5/38.8 MB 7.9 MB/s eta
0:00:01
----- 38.8/38.8 MB 7.6 MB/s
0:00:05
Downloading pi_heif-1.3.0-cp311-cp311-win_amd64.whl (1.9 MB)
----- 0.0/1.9 MB ? eta -:--:--
----- 1.9/1.9 MB 9.0 MB/s
0:00:00
Downloading pillow_avif_plugin-1.5.5-cp311-cp311-win_amd64.whl (9.9
MB)
----- 0.0/9.9 MB ? eta -:--:--
----- 2.4/9.9 MB 11.2 MB/s eta
0:00:01
----- 2.9/9.9 MB 7.3 MB/s eta
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----- 3.9/9.9 MB 5.9 MB/s eta
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----- 5.8/9.9 MB 6.5 MB/s eta
0:00:01
----- 6.3/9.9 MB 6.7 MB/s eta
0:00:01
----- 6.8/9.9 MB 5.1 MB/s eta
0:00:01
----- 7.1/9.9 MB 4.8 MB/s eta
0:00:01
----- 8.7/9.9 MB 4.9 MB/s eta
0:00:01
----- 9.9/9.9 MB 5.1 MB/s
0:00:01
Downloading tqdm-4.67.3-py3-none-any.whl (78 kB)
Downloading filetype-1.2.0-py2.py3-none-any.whl (19 kB)
Downloading python_dotenv-1.2.2-py3-none-any.whl (22 kB)
Downloading requests_toolbelt-1.0.0-py2.py3-none-any.whl (54 kB)
Installing collected packages: pillow-avif-plugin, filetype, tqdm,
python-dotenv, pi-heif, opencv-python-headless, idna, requests-
toolbelt, roboflow
----- 0/9 [pillow-avif-plugin]
----- 0/9 [pillow-avif-plugin]
----- 1/9 [filetype]
----- 1/9 [filetype]

```



```

----- 6/9 [idna]
----- 6/9 [idna]
----- 7/9 [requests-toolbelt]
----- 7/9 [requests-toolbelt]
----- 7/9 [requests-toolbelt]
----- 7/9 [requests-toolbelt]
----- 8/9 [roboflow]
----- 8/9 [roboflow]
----- 8/9 [roboflow]
----- 8/9 [roboflow]
----- 8/9 [roboflow]
----- 8/9 [roboflow]
----- 9/9 [roboflow]

```

Successfully installed filetype-1.2.0 idna-3.7 opencv-python-headless-4.10.0.84 pi-heif-1.3.0 pillow-avif-plugin-1.5.5 python-dotenv-1.2.2 requests-toolbelt-1.0.0 roboflow-1.2.16 tqdm-4.67.3

Note: you may need to restart the kernel to use updated packages.

WARNING: Ignoring invalid distribution ~atplotlib (c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages)

WARNING: Ignoring invalid distribution ~atplotlib (c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages)

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

tensorflow 2.21.0 requires keras>=3.12.0, but you have keras 2.15.0 which is incompatible.

tensorflow 2.21.0 requires ml_dtypes<1.0.0,>=0.5.1, but you have ml_dtypes 0.2.0 which is incompatible.

tensorflow 2.21.0 requires protobuf<8.0.0,>=6.31.1, but you have protobuf 4.25.8 which is incompatible.

WARNING: Ignoring invalid distribution ~atplotlib (c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages)

Requirement already satisfied: matplotlib in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (3.10.8)

Requirement already satisfied: contourpy>=1.0.1 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib) (1.3.3)

Requirement already satisfied: cyclor>=0.10 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib) (4.62.1)

Requirement already satisfied: kiwisolver>=1.3.1 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib) (1.5.0)

Requirement already satisfied: numpy>=1.23 in c:\Users\cyanezru\

AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib) (2.4.3)
Requirement already satisfied: packaging>=20.0 in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from matplotlib) (26.0)
Requirement already satisfied: pillow>=8 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib) (12.1.1)
Requirement already satisfied: pyparsing>=3 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
Note: you may need to restart the kernel to use updated packages.

WARNING: Ignoring invalid distribution ~atplotlib (c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages)
WARNING: Ignoring invalid distribution ~atplotlib (c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages)
WARNING: Ignoring invalid distribution ~atplotlib (c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages)

Collecting numpy==1.26.4
Using cached numpy-1.26.4-cp311-cp311-win_amd64.whl.metadata (61 kB)
Using cached numpy-1.26.4-cp311-cp311-win_amd64.whl (15.8 MB)
Installing collected packages: numpy
Attempting uninstall: numpy
Found existing installation: numpy 2.4.3
Uninstalling numpy-2.4.3:
Successfully uninstalled numpy-2.4.3
Successfully installed numpy-1.26.4
Note: you may need to restart the kernel to use updated packages.

WARNING: Ignoring invalid distribution ~atplotlib (c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages)
WARNING: Ignoring invalid distribution ~atplotlib (c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages)
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.
opencv-python 4.13.0.92 requires numpy>=2; python_version >= "3.9", but you have numpy 1.26.4 which is incompatible.
tensorflow 2.21.0 requires keras>=3.12.0, but you have keras 2.15.0 which is incompatible.
tensorflow 2.21.0 requires ml_dtypes<1.0.0,>=0.5.1, but you have ml_dtypes 0.2.0 which is incompatible.


```
tensorflow 2.21.0 requires protobuf<8.0.0,>=6.31.1, but you have
protobuf 4.25.8 which is incompatible.
WARNING: Ignoring invalid distribution ~atplotlib (c:\Users\cyanezru\
AppData\Local\Programs\Python\Python311\Lib\site-packages)
```

Step 2: Import Required Libraries

```
import cv2
from roboflow import Roboflow
from ultralytics import YOLO
import matplotlib.pyplot as plt
import numpy as np
import os
from IPython.display import Image, display
```

Creating new Ultralytics Settings v0.0.6 file
View Ultralytics Settings with 'yolo settings' or at 'C:\Users\cyanezru\AppData\Roaming\Ultralytics\settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see <https://docs.ultralytics.com/quickstart/#ultralytics-settings>.

Step 3: Set Up RoboFlow Account

1. Go to RoboFlow and create a free account
2. Create a new project and select "Object Detection" as project type
3. Choose "YOLOv26" as the model format

Step 4: Upload and Annotate Your Images

Using RoboFlow Web Interface:

1. Click "Upload" to add your images
2. Use the annotation tool to draw bounding boxes around objects
3. Assign the correct class labels to each bounding box
4. Apply preprocessing and augmentation as needed

Step 5. Download your Custom your Dataset

1. In RoboFlow, go to your project dashboard
2. Click "Generate" to create a dataset version
3. Apply any desired augmentations
4. Click "Export" and select "YOLOv~~x~~" format (x the version you are using)
5. Copy the download code

```
# Initialize RoboFlow (you'll need your API key)
```

```
!pip install roboflow
```

```
from roboflow import Roboflow
```

```
rf = Roboflow(api_key="JveMfT08Up00ztjKycz2")
project = rf.workspace("carlosyanezs-
workspace").project("clase3_grupo3")
version = project.version(1)
dataset = version.download("yolo26")
```

Requirement already satisfied: roboflow in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (1.2.16)

Requirement already satisfied: certifi in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (2026.2.25)

Requirement already satisfied: idna==3.7 in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (3.7)

Requirement already satisfied: cyciler in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (0.12.1)

Requirement already satisfied: kiwisolver>=1.3.1 in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (1.5.0)

Requirement already satisfied: matplotlib in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (3.10.8)

Requirement already satisfied: numpy>=1.18.5 in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (1.26.4)

Requirement already satisfied: opencv-python-headless==4.10.0.84 in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (4.10.0.84)

Requirement already satisfied: Pillow>=7.1.2 in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (12.1.1)

Requirement already satisfied: pillow-avif-plugin<2 in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (1.5.5)

Requirement already satisfied: python-dateutil in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from roboflow) (2.9.0.post0)

Requirement already satisfied: python-dotenv in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (1.2.2)

Requirement already satisfied: requests in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (2.32.5)

Requirement already satisfied: six in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from roboflow) (1.17.0)

Requirement already satisfied: urllib3>=1.26.6 in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (2.6.3)

Requirement already satisfied: tqdm>=4.41.0 in C:\Users\cyanezru\

AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (4.67.3)
Requirement already satisfied: PyYAML<=5.3.1 in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (6.0.3)
Requirement already satisfied: requests-toolbelt in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (1.0.0)
Requirement already satisfied: filetype in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (1.2.0)
Requirement already satisfied: pi-heif<2 in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (1.3.0)
Requirement already satisfied: colorama in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from tqdm<=4.41.0->roboflow) (0.4.6)
Requirement already satisfied: contourpy<=1.0.1 in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib->roboflow) (1.3.3)
Requirement already satisfied: fonttools<=4.22.0 in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib->roboflow) (4.62.1)
Requirement already satisfied: packaging<=20.0 in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from matplotlib->roboflow) (26.0)
Requirement already satisfied: pyparsing<=3 in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib->roboflow) (3.3.2)
Requirement already satisfied: charset-normalizer<4,>=2 in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from requests->roboflow) (3.4.6)

WARNING: Ignoring invalid distribution ~atplotlib (C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages)

WARNING: Ignoring invalid distribution ~atplotlib (C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages)

WARNING: Ignoring invalid distribution ~atplotlib (C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages)

loading Roboflow workspace...

loading Roboflow project...

Downloading Dataset Version Zip in Clase3_Grupo3-1 to yolo26:: 100%|
██████████| 1938/1938 [00:00<00:00, 2162.06it/s]

Extracting Dataset Version Zip to Clase3_Grupo3-1 in yolo26:: 100%|
██████████| 119/119 [00:00<00:00, 272.80it/s]

Step 6: Train Your Custom YOLO Model

```
# Load the YOLO model with custom configuration
model = YOLO('yolo26n.pt') # Using nano version as base

# Train the model on your custom dataset
results = model.train(
    data=f'{dataset.location}/data.yaml', # Path to dataset config
    epochs=20, # Number of training epochs
    imgsz=640, # Image size
    batch=16, # Batch size
    name='custom_yolo_model' # Name for the training run
)

/ultralytics/assets/releases/download/v8.4.0/yolo26n.pt to
'yolo26n.pt': 100% _____ 5.3MB 5.2MB/s 1.0s.9s<0.2s1.9ss
Ultralytics 8.4.26 Python-3.11.9 torch-2.11.0+cpu CPU (13th Gen Intel
Core i7-1355U)
engine\trainer: agnostic_nms=False, amp=True, angle=1.0,
augment=False, auto_augment=randaugument, batch=16, bgr=0.0, box=7.5,
cache=False, cfg=None, classes=None, close_mosaic=10, cls=0.5,
compile=False, conf=None, copy_paste=0.0, copy_paste_mode=flip,
cos_lr=False, cutmix=0.0, data=c:\Users\cyanezru\OneDrive - NTT DATA
EMEAL\Documentos\Repositorios\Vision-por-computador-MIA-B---Grupo-3\
Clase_3\Clase3_Grupo3-1\data.yaml, degrees=0.0, deterministic=True,
device=cpu, dfl=1.5, dnn=False, dropout=0.0, dynamic=False,
embed=None, end2end=None, epochs=20, erasing=0.4, exist_ok=False,
fliplr=0.5, flipud=0.0, format=torchscript, fraction=1.0, freeze=None,
half=False, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, imgsz=640, int8=False,
iou=0.7, keras=False, kobj=1.0, line_width=None, lr0=0.01, lrf=0.01,
mask_ratio=4, max_det=300, mixup=0.0, mode=train, model=yolo26n.pt,
momentum=0.937, mosaic=1.0, multi_scale=0.0, name=custom_yolo_model,
nbs=64, nms=False, opset=None, optimize=False, optimizer=auto,
overlap_mask=True, patience=100, perspective=0.0, plots=True,
pose=12.0, pretrained=True, profile=False, project=None, rect=False,
resume=False, retina_masks=False, rle=1.0, save=True, save_conf=False,
save_crop=False, save_dir=C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\
Documentos\Repositorios\Vision-por-computador-MIA-B---Grupo-3\Clase_3\
runs\detect\custom_yolo_model, save_frames=False, save_json=False,
save_period=-1, save_txt=False, scale=0.5, seed=0, shear=0.0,
show=False, show_boxes=True, show_conf=True, show_labels=True,
simplify=True, single_cls=False, source=None, split=val,
stream_buffer=False, task=detect, time=None, tracker=botsort.yaml,
translate=0.1, val=True, verbose=True, vid_stride=1, visualize=False,
warmup_bias_lr=0.1, warmup_epochs=3.0, warmup_momentum=0.8,
weight_decay=0.0005, workers=8, workspace=None
/assets/Arial.ttf to 'C:\Users\cyanezru\AppData\Roaming\Ultralytics\
Arial.ttf': 100% _____ 755.1KB 2.4MB/s 0.3s 0.2s<0.2s
Overriding model.yaml nc=80 with nc=2
```

	from	n	params	module
arguments				
0	-1	1	464	ultralytics.nn.modules.conv.Conv
[3, 16, 3, 2]				
1	-1	1	4672	ultralytics.nn.modules.conv.Conv
[16, 32, 3, 2]				
2	-1	1	6640	ultralytics.nn.modules.block.C3k2
				[32, 64, 1, False, 0.25]
3	-1	1	36992	ultralytics.nn.modules.conv.Conv
[64, 64, 3, 2]				
4	-1	1	26080	ultralytics.nn.modules.block.C3k2
				[64, 128, 1, False, 0.25]
5	-1	1	147712	ultralytics.nn.modules.conv.Conv
[128, 128, 3, 2]				
6	-1	1	87040	ultralytics.nn.modules.block.C3k2
				[128, 128, 1, True]
7	-1	1	295424	ultralytics.nn.modules.conv.Conv
[128, 256, 3, 2]				
8	-1	1	346112	ultralytics.nn.modules.block.C3k2
				[256, 256, 1, True]
9	-1	1	164608	ultralytics.nn.modules.block.SPPF
				[256, 256, 5, 3, True]
10	-1	1	249728	ultralytics.nn.modules.block.C2PSA
				[256, 256, 1]
11	-1	1	0	torch.nn.modules.upsampling.Upsample
				[None, 2, 'nearest']
12	[-1, 6]	1	0	ultralytics.nn.modules.conv.Concat
				[1]
13	-1	1	119808	ultralytics.nn.modules.block.C3k2
				[384, 128, 1, True]
14	-1	1	0	torch.nn.modules.upsampling.Upsample
				[None, 2, 'nearest']
15	[-1, 4]	1	0	ultralytics.nn.modules.conv.Concat
				[1]
16	-1	1	34304	ultralytics.nn.modules.block.C3k2
				[256, 64, 1, True]
17	-1	1	36992	ultralytics.nn.modules.conv.Conv
[64, 64, 3, 2]				

```

18          [-1, 13] 1      0
ultralytics.nn.modules.conv.Concat          [1]

19          -1 1      95232
ultralytics.nn.modules.block.C3k2          [192, 128, 1, True]

20          -1 1      147712 ultralytics.nn.modules.conv.Conv
[128, 128, 3, 2]
21          [-1, 10] 1      0
ultralytics.nn.modules.conv.Concat          [1]

22          -1 1      463104
ultralytics.nn.modules.block.C3k2          [384, 256, 1, True, 0.5,
True]
23          [16, 19, 22] 1      241956
ultralytics.nn.modules.head.Detect          [2, 1, True, [64, 128,
256]]
YOLO26n summary: 260 layers, 2,504,580 parameters, 2,504,580
gradients, 5.8 GFLOPs

```

```

Transferred 606/708 items from pretrained weights
train: Fast image access (ping: 0.20.0 ms, read: 100.937.3 MB/s,
size: 40.3 KB)
train: Scanning C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\
Documentos\Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\
Clase3_Grupos-1\train\labels... 27 images, 0 backgrounds, 0 corrupt:
100% _____ 27/27 656.3it/s 0.0s
train: New cache created: C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\
Documentos\Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\
Clase3_Grupos-1\train\labels.cache
val: Fast image access (ping: 0.20.0 ms, read: 82.416.6 MB/s, size:
35.7 KB)
val: Scanning C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\Documentos\
Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\
Clase3_Grupos-1\valid\labels... 21 images, 0 backgrounds, 0 corrupt:
100% _____ 21/21 1.4Kit/s 0.0s
val: New cache created: C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\
Documentos\Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\
Clase3_Grupos-1\valid\labels.cache
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and
'momentum=0.937' and determining best 'optimizer', 'lr0' and
'momentum' automatically...
optimizer: AdamW(lr=0.001667, momentum=0.9) with parameter groups 114
weight(decay=0.0), 126 weight(decay=0.0005), 126 bias(decay=0.0)
Plotting labels to C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\
Documentos\Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\
runs\detect\custom_yolo_model\labels.jpg...
Image sizes 640 train, 640 val
Using 0 dataloader workers
Logging results to C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\

```

Documentos\Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\
 runs\detect\custom_yolo_model
 Starting training for 20 epochs...

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
ages	Instances	Box(P	R	mAP50	mAP50-95): 100%
	1/1	4.3s/it 4.3s			
		all	21	21	0.00254 0.75
0.0507	0.0289				

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
ages	Instances	Box(P	R	mAP50	mAP50-95): 100%
	1/1	3.4s/it 3.4s			
		all	21	21	0.00253 0.75
0.0643	0.0369				

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
ages	Instances	Box(P	R	mAP50	mAP50-95): 100%
	1/1	2.8s/it 2.8s			
		all	21	21	0.0031 0.875
0.0681	0.0394				

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
ages	Instances	Box(P	R	mAP50	mAP50-95): 100%
	1/1	2.5s/it 2.5s			
		all	21	21	0.00353 0.875
0.0616	0.0384				

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
ages	Instances	Box(P	R	mAP50	mAP50-95): 100%
	1/1	2.9s/it 2.9s			
		all	21	21	0.00411 0.875
0.0479	0.0333				

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
ages	Instances	Box(P	R	mAP50	mAP50-95): 100%
	1/1	2.9s/it 2.9s			
		all	21	21	0.00482 0.938
0.0473	0.0348				

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
ages	Instances	Box(P	R	mAP50	mAP50-95): 100%
	1/1	2.9s/it 2.9s			

0.0577	0.0442	all	21	21	0.00529	0.938
--------	--------	-----	----	----	---------	-------

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
ages	Instances	Box(P	R	mAP50	mAP50-95): 100%
	1/1	2.9s/it	2.9s		

		all	21	21	0.00559	0.938
0.075	0.0578					

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
ages	Instances	Box(P	R	mAP50	mAP50-95): 100%
	1/1	2.9s/it	2.9s		

		all	21	21	0.00561	0.938
0.0755	0.0595					

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
ages	Instances	Box(P	R	mAP50	mAP50-95): 100%
	1/1	2.9s/it	2.9s		

		all	21	21	0.00576	0.938
0.0795	0.0635					

Closing dataloader mosaic

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
ages	Instances	Box(P	R	mAP50	mAP50-95): 100%
	1/1	3.2s/it	3.2s		

		all	21	21	0.00574	0.938
0.093	0.0728					

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
ages	Instances	Box(P	R	mAP50	mAP50-95): 100%
	1/1	3.0s/it	3.0s		

		all	21	21	0.00565	0.938
0.0914	0.0708					

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
ages	Instances	Box(P	R	mAP50	mAP50-95): 100%
	1/1	3.2s/it	3.2s		

		all	21	21	0.00557	0.938
0.099	0.0769					

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
ages	Instances	Box(P	R	mAP50	mAP50-95): 100%
	1/1	2.7s/it	2.7s		


```
all 21 21 0.00536 0.938
0.104 0.0783
```

```
Epoch GPU_mem box_loss cls_loss dfl_loss Instances
Size ages Instances Box(P R mAP50 mAP50-95): 100%
----- 1/1 2.9s/it 2.9s
all 21 21 0.00528 0.938
0.103 0.0778
```

```
Epoch GPU_mem box_loss cls_loss dfl_loss Instances
Size ages Instances Box(P R mAP50 mAP50-95): 100%
----- 1/1 3.0s/it 3.0s
all 21 21 0.00515 0.938
0.109 0.0813
```

```
Epoch GPU_mem box_loss cls_loss dfl_loss Instances
Size ages Instances Box(P R mAP50 mAP50-95): 100%
----- 1/1 3.8s/it 3.8s
all 21 21 0.00508 0.938
0.116 0.0848
```

```
Epoch GPU_mem box_loss cls_loss dfl_loss Instances
Size ages Instances Box(P R mAP50 mAP50-95): 100%
----- 1/1 4.2s/it 4.2s
all 21 21 0.00508 0.938
0.151 0.109
```

```
Epoch GPU_mem box_loss cls_loss dfl_loss Instances
Size ages Instances Box(P R mAP50 mAP50-95): 100%
----- 1/1 3.5s/it 3.5s
all 21 21 0.00501 0.938
0.184 0.136
```

```
Epoch GPU_mem box_loss cls_loss dfl_loss Instances
Size ages Instances Box(P R mAP50 mAP50-95): 100%
----- 1/1 3.0s/it 3.0s
all 21 21 0.00497 0.938
0.185 0.137
```

20 epochs completed in 0.092 hours.
Optimizer stripped from C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\Documentos\Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\runs\detect\custom_yolo_model\weights\last.pt, 5.4MB
Optimizer stripped from C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\

```
Documentos\Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\
runs\detect\custom_yolo_model\weights\best.pt, 5.4MB
```

```
Validating C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\Documentos\
Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\runs\
detect\custom_yolo_model\weights\best.pt...
```

```
Ultralytics 8.4.26 Python-3.11.9 torch-2.11.0+cpu CPU (13th Gen Intel
Core i7-1355U)
```

```
YOLO26n summary (fused): 122 layers, 2,375,226 parameters, 0
gradients, 5.2 GFLOPs
```

ages	Instances	Box(P	R	mAP50	mAP50-95):
	1/1	2.3s/it	2.3s		100%
		all	21	21	0.00497 0.938
0.185	0.137	oximeter	8	8	0.00147 0.875
0.158	0.122	pencil_sharpener	13	13	0.00846 1
0.211	0.151				

```
Speed: 2.6ms preprocess, 85.6ms inference, 0.0ms loss, 0.1ms
postprocess per image
```

```
Results saved to C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\
Documentos\Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\
runs\detect\custom_yolo_model
```

Step 7: Evaluate Your Model

```
# Evaluate model performance
metrics = model.val() # Validate the model
print(f"mAP50-95: {metrics.box.map}")
print(f"mAP50: {metrics.box.map50}")
```

```
# Show some validation results
!yolo task=detect mode=val
model=runs/detect/custom_yolo_model/weights/best.pt
data={dataset.location}/data.yaml
```

```
Ultralytics 8.4.26 Python-3.11.9 torch-2.11.0+cpu CPU (13th Gen Intel
Core i7-1355U)
```

```
YOLO26n summary (fused): 122 layers, 2,375,226 parameters, 0
gradients, 5.2 GFLOPs
```

```
val: Fast image access (ping: 0.20.0 ms, read: 72.834.7 MB/s, size:
36.4 KB)
```

```
val: Scanning C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\Documentos\
Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\
Clase3_Grupo3-1\valid\labels.cache... 21 images, 0 backgrounds, 0
corrupt: 100%
```

ages	Instances	Box(P	R	mAP50	mAP50-95):
	2/2	1.1s/it	2.3s5.9s		100%
		all	21	21	0.00497 0.938
0.185	0.137				

	oximeter	8	8	0.00147	0.875
0.158	0.122				
	pencil_sharpener	13	13	0.00846	1
0.211	0.151				

Speed: 2.8ms preprocess, 94.6ms inference, 0.0ms loss, 0.1ms postprocess per image
Results saved to C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\Documentos\Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\runs\detect\val
mAP50-95: 0.13665421974490494
mAP50: 0.1845619323430609

```
Traceback (most recent call last):
  File "<frozen runpy>", line 198, in _run_module_as_main
  File "<frozen runpy>", line 88, in _run_code
  File "C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Scripts\yolo.exe\__main__.py", line 5, in <module>
  File "c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages\ultralytics\cfg\__init__.py", line 907, in entrypoint
    check_dict_alignment(full_args_dict, {a: ""})
  File "c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages\ultralytics\cfg\__init__.py", line 506, in check_dict_alignment
    raise SyntaxError(string + CLI_HELP_MSG) from e
SyntaxError: '-' is not a valid YOLO argument.
```

Arguments received: ['yolo', 'task=detect', 'mode=val', 'model=runs/detect/custom_yolo_model/weights/best.pt', 'data=c:\\Users\\cyanezru\\OneDrive', '-', 'NTT', 'DATA', 'EMEAL\\Documentos\\Repositories\\Vision-por-computador-MIA-B---Grupo-3\\Clase_3\\Clase3_Grupo3-1\data.yaml']. Ultralytics 'yolo' commands use the following syntax:

yolo TASK MODE ARGS

Where TASK (optional) is one of ['classify', 'pose', 'segment', 'obb', 'detect']
MODE (required) is one of ['track', 'export', 'benchmark', 'train', 'val', 'predict']
ARGS (optional) are any number of custom 'arg=value' pairs like 'imgsz=320' that override defaults.
See all ARGS at <https://docs.ultralytics.com/usage/cfg> or with 'yolo cfg'

1. Train a detection model for 10 epochs with an initial learning_rate of 0.01
yolo train data=coco8.yaml model=yolo26n.pt epochs=10 lr0=0.01

2. Predict a YouTube video using a pretrained segmentation model at image size 320:

```
yolo predict model=yolo26n-seg.pt
source='https://youtu.be/LNw0DJXcvt4' imgsz=320
```

3. Validate a pretrained detection model at batch-size 1 and image size 640:

```
yolo val model=yolo26n.pt data=coco8.yaml batch=1 imgsz=640
```

4. Export a YOLO26n classification model to ONNX format at image size 224 by 128 (no TASK required)

```
yolo export model=yolo26n-cls.pt format=onnx imgsz=224,128
```

5. Ultralytics solutions usage

```
yolo solutions count or any of ['crop', 'blur', 'workout',
'heatmap', 'isegment', 'visioneye', 'speed', 'queue', 'analytics',
'inference', 'trackzone'] source="path/to/video.mp4"
```

6. Run special commands:

```
yolo help
yolo checks
yolo version
yolo settings
yolo copy-cfg
yolo cfg
yolo solutions help
```

Docs: <https://docs.ultralytics.com>

Solutions: <https://docs.ultralytics.com/solutions/>

Community: <https://community.ultralytics.com>

GitHub: <https://github.com/ultralytics/ultralytics>

Step 8: Test Your Custom Model on an Image

```
# Load your custom trained model
```

```
custom_model = YOLO(r'C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\
Documentos\Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\
runs\detect\custom_yolo_model\weights\best.pt')
```

```
# Test on a sample image
```

```
results = custom_model(r'C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\
Documentos\Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\
dataset\oximeter.jpg', conf=0.25)
```

```
# Display results
```

```
%matplotlib inline
```

```
for result in results:
```

```
    img = result.plot() # Plot detections
```

```
    img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
```

```
    plt.figure(figsize=(12, 8))
```

```
    plt.imshow(img_rgb)
```

```
plt.axis('off')  
plt.show()
```

image 1/1 C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\Documentos\Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\dataset\oximeter.jpg: 640x640 (no detections), 264.7ms
Speed: 18.7ms preprocess, 264.7ms inference, 0.3ms postprocess per image at shape (1, 3, 640, 640)



Step 9: Test Your Custom Model on Video

```
# Write your code here..
```

Activity: Create Your Custom Detector

1. **Create a RoboFlow account** and set up a new project
2. **Upload at least 25 images** of your custom objects (at least 2)
3. **Annotate all images** with bounding boxes and class labels
4. **Train a YOLO model** on your custom dataset
5. **Test your model** on new images and evaluate its performance
6. **Implement real-time detection** with your custom classes

Grading Rubric:

Criteria	Points	Description
1. Dataset Creation	4	Successfully creates and annotates a custom dataset with at least 25 images and multiple classes in RoboFlow.
2. Model Training	1	Properly trains a YOLOv8 model on the custom dataset with appropriate hyperparameters.
3. Model Evaluation	1	Evaluates model performance with appropriate metrics and provides analysis of results.
4. Custom Detection	3	Implements real-time detection with custom classes and displays results correctly. You have to use video from your webcam, no images or saved videos allowed.
5. Documentation	1	Provides clear documentation of the process, including dataset details and model performance.

Total: 10 points

Tips for Success:

- **Quality over quantity:** Well-annotated images are more important than many poorly annotated ones
- **Class balance:** Try to have a similar number of examples for each class
- **Variety:** Include different angles, lighting conditions, and backgrounds
- **Augmentation:** Use RoboFlow's augmentation features to expand your dataset